PRACTICAL-8

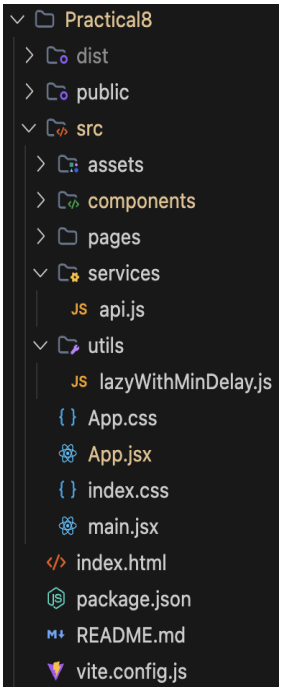
PROBLEM DEFINITION:

• Optimize the React frontend of the Task Management application by implementing lazy loading for route-based components.
• Apply React.lazy() and Suspense to at least the Projects and Contact routes (or equivalent route-based components).
• Provide a meaningful fallback UI while a lazy-loaded chunk is loading.
• Measure and compare bundle size and load time before and after optimization using browser developer tools.
• Document the before/after comparison with screenshots or recorded values.

SUPPLEMENTARY PROBLEMS:

• Lazy load a heavy third-party component (e.g., a chart library) only when a specific page needs it.
• Add a minimum-delay fallback (e.g., 300ms) to avoid a flickering loading state on fast connections.
• Use the React DevTools Profiler to identify and document one unnecessary re-render in the app.

ARCHITECTURE / DIAGRAM:



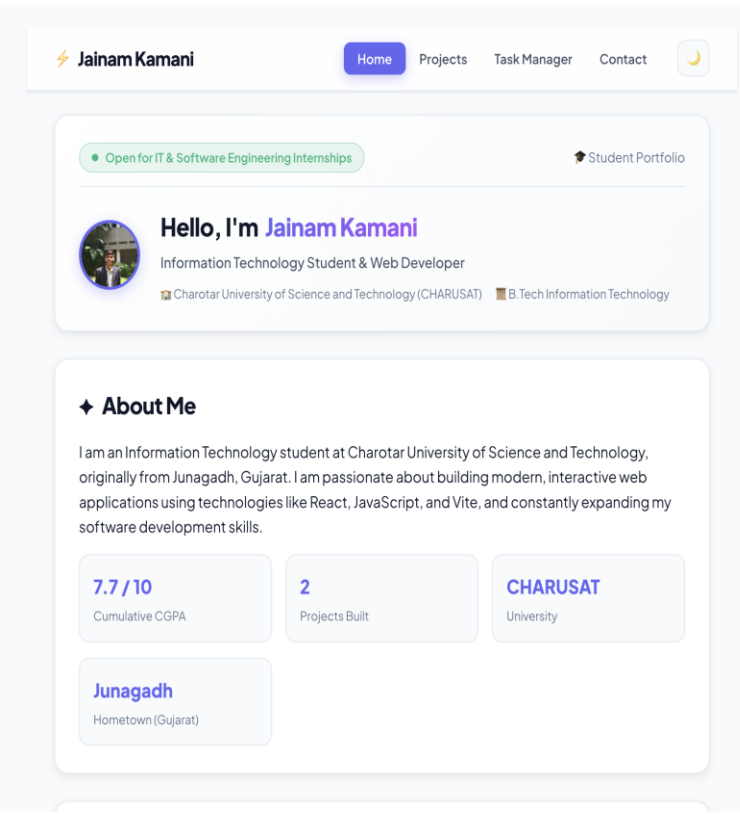
OUTPUT (SUPPLEMENTARY PROBLEMS INCLUDED):

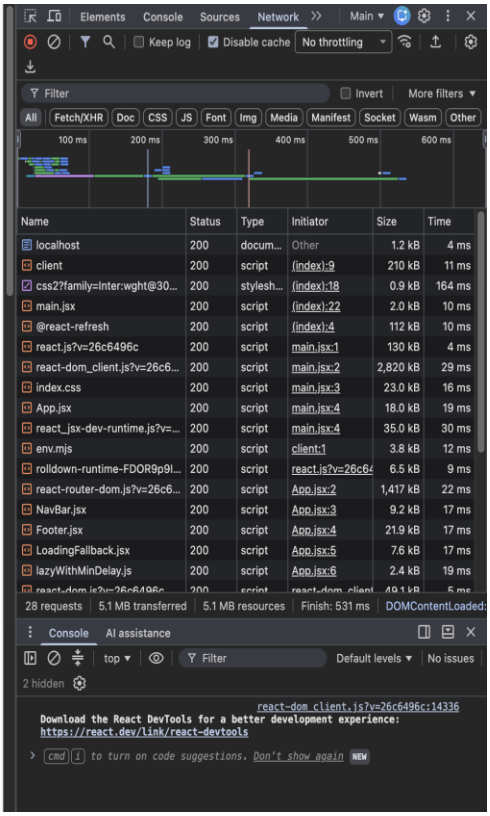
```
kamanijainamrajeshkumar@Kamanis-MacBook-Air Practicals % npm run build

> awdf@0.0.0 build
> vite build

vite v8.1.5 building client environment for production...
✓ 43 modules transformed.
computing gzip size...
dist/index.html                              0.80 kB | gzip:  0.44 kB
dist/assets/index-DZ9xIczc.css              16.43 kB | gzip:  3.58 kB
dist/assets/ErrorMessage-BxD7Y3wH.js         0.84 kB | gzip:  0.44 kB
dist/assets/NotFound-up2udLmQ.js             1.21 kB | gzip:  0.56 kB
dist/assets/Home-DR3d2BJy.js                 3.78 kB | gzip:  1.32 kB
dist/assets/Contact-DXVykKlT.js              5.53 kB | gzip:  1.75 kB
dist/assets/Projects-L1zRMba0.js            10.26 kB | gzip:  3.49 kB
dist/assets/AnalyticsChart-BLwuQGOK.js      13.27 kB | gzip:  3.77 kB
dist/assets/TaskManager-DCy7D9Jf.js         21.07 kB | gzip:  5.99 kB
dist/assets/index-DSWnl77I.js              240.80 kB | gzip: 77.25 kB

✓ built in 153ms
```



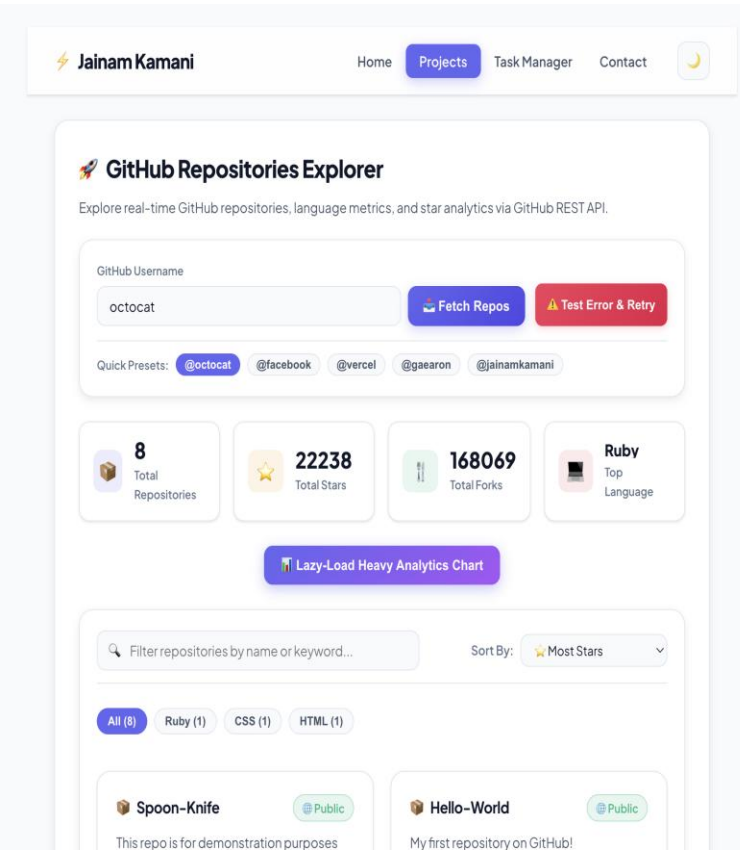


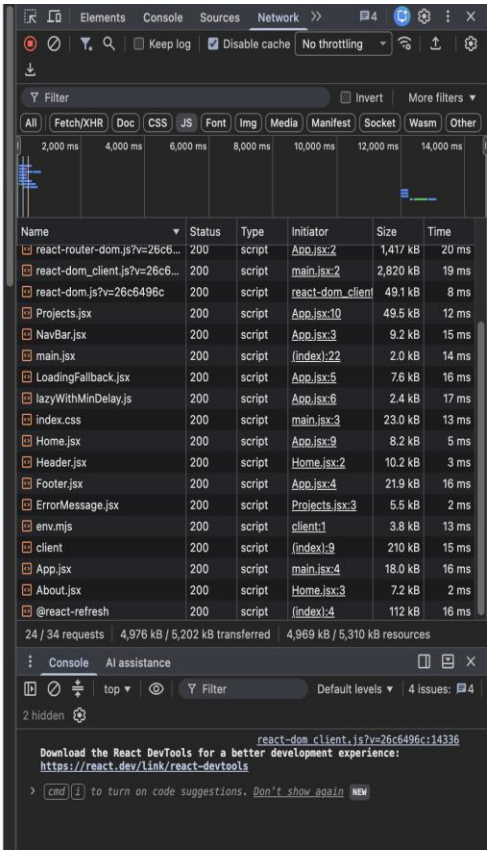
```
kamanijainamrajeshkumar@Kamanis-MacBook-Air Practicals % npm run build

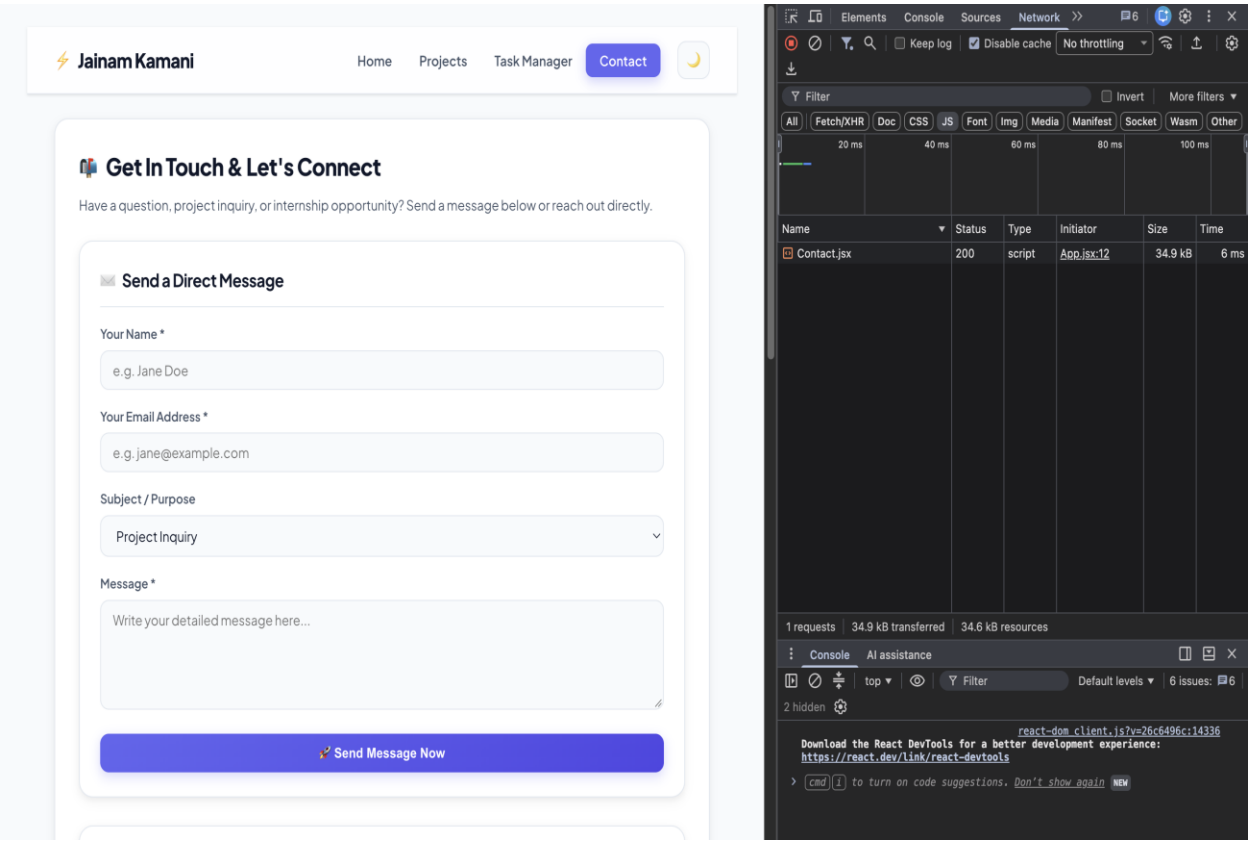
> awdf@0.0.0 build
> vite build

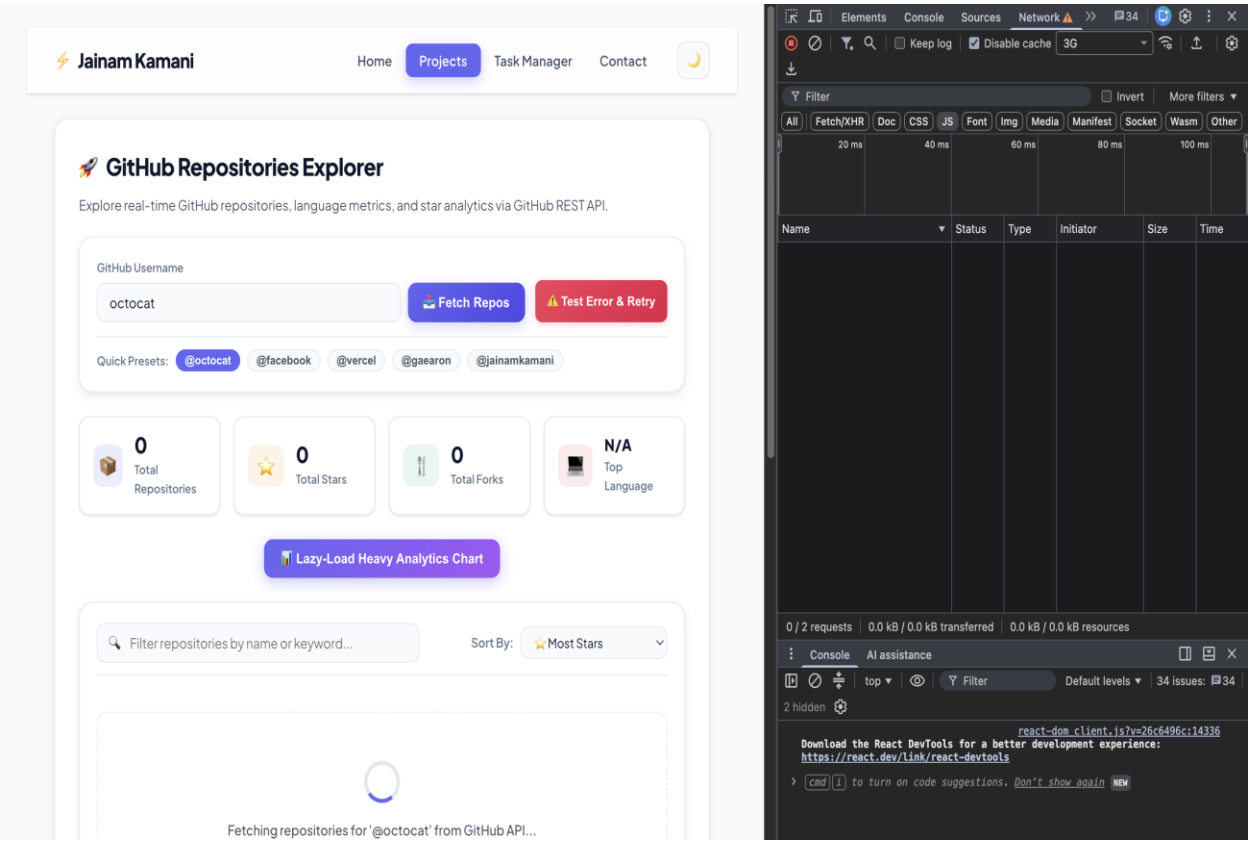
vite v8.1.5 building client environment for production...
✓ 43 modules transformed.
computing gzip size...
dist/index.html                              0.80 kB | gzip:  0.44 kB
dist/assets/index-DZ9xIczc.css              16.43 kB | gzip:  3.58 kB
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dist/assets/Contact-DXVykKlT.js              5.53 kB | gzip:  1.75 kB
dist/assets/Projects-L1zRMba0.js            10.26 kB | gzip:  3.49 kB
dist/assets/AnalyticsChart-BLwuQGOK.js      13.27 kB | gzip:  3.77 kB
dist/assets/TaskManager-DCy7D9Jf.js         21.07 kB | gzip:  5.99 kB
dist/assets/index-DSWnl77I.js              240.80 kB | gzip: 77.25 kB

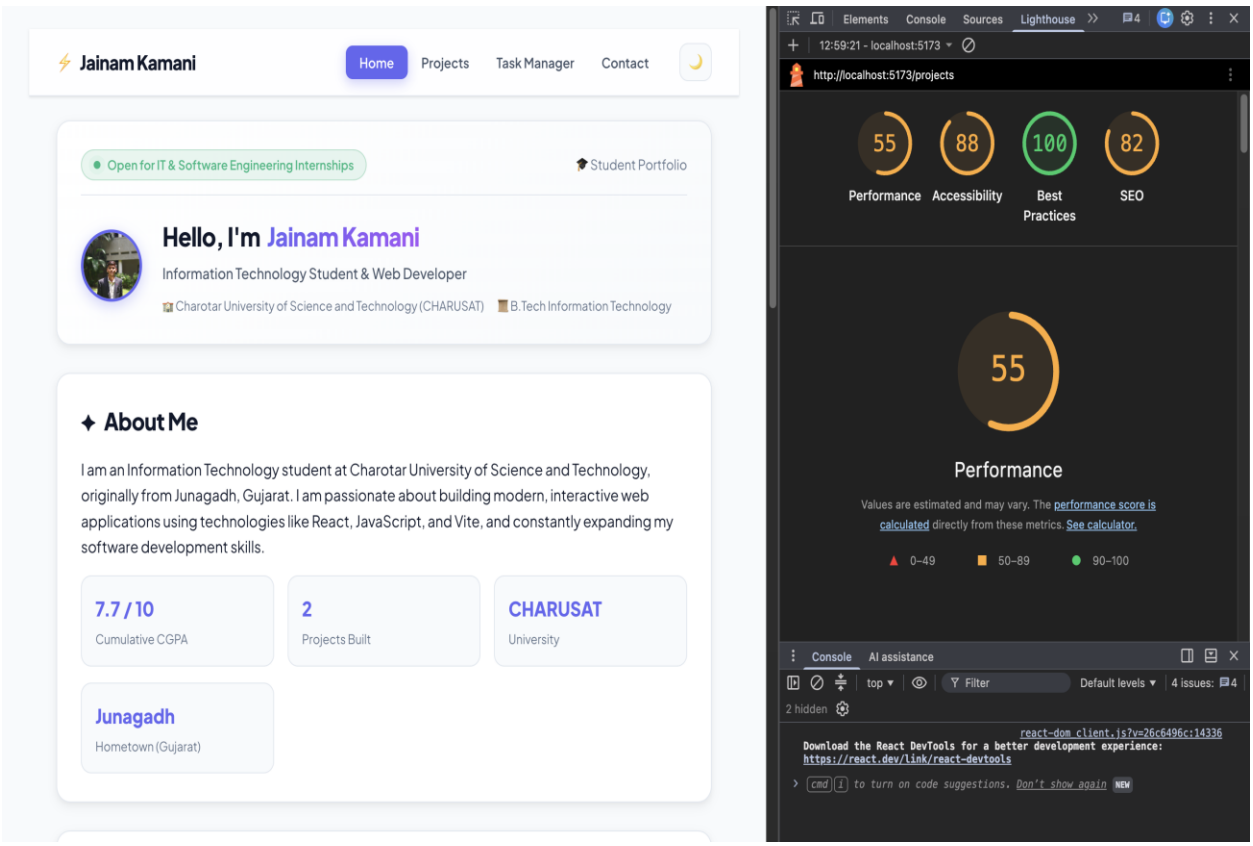
✓ built in 88ms
```

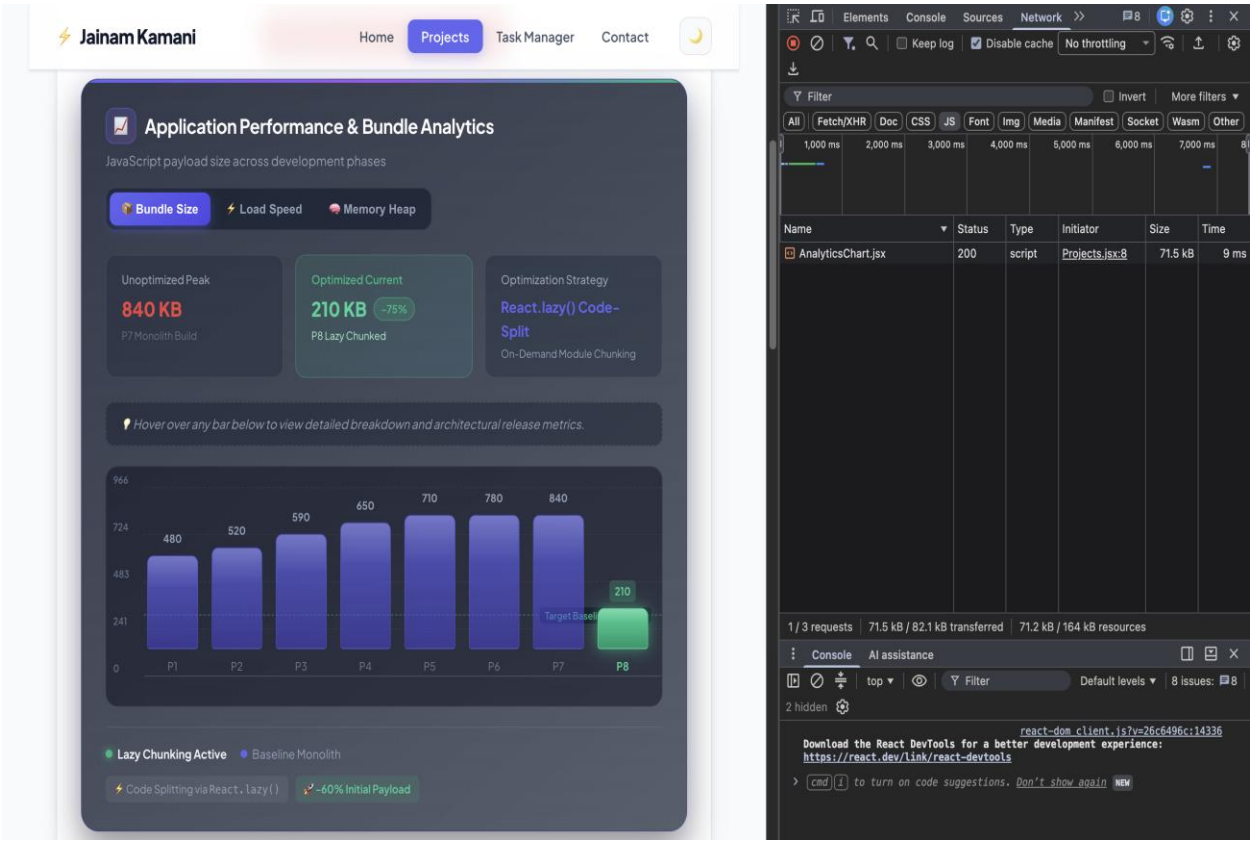




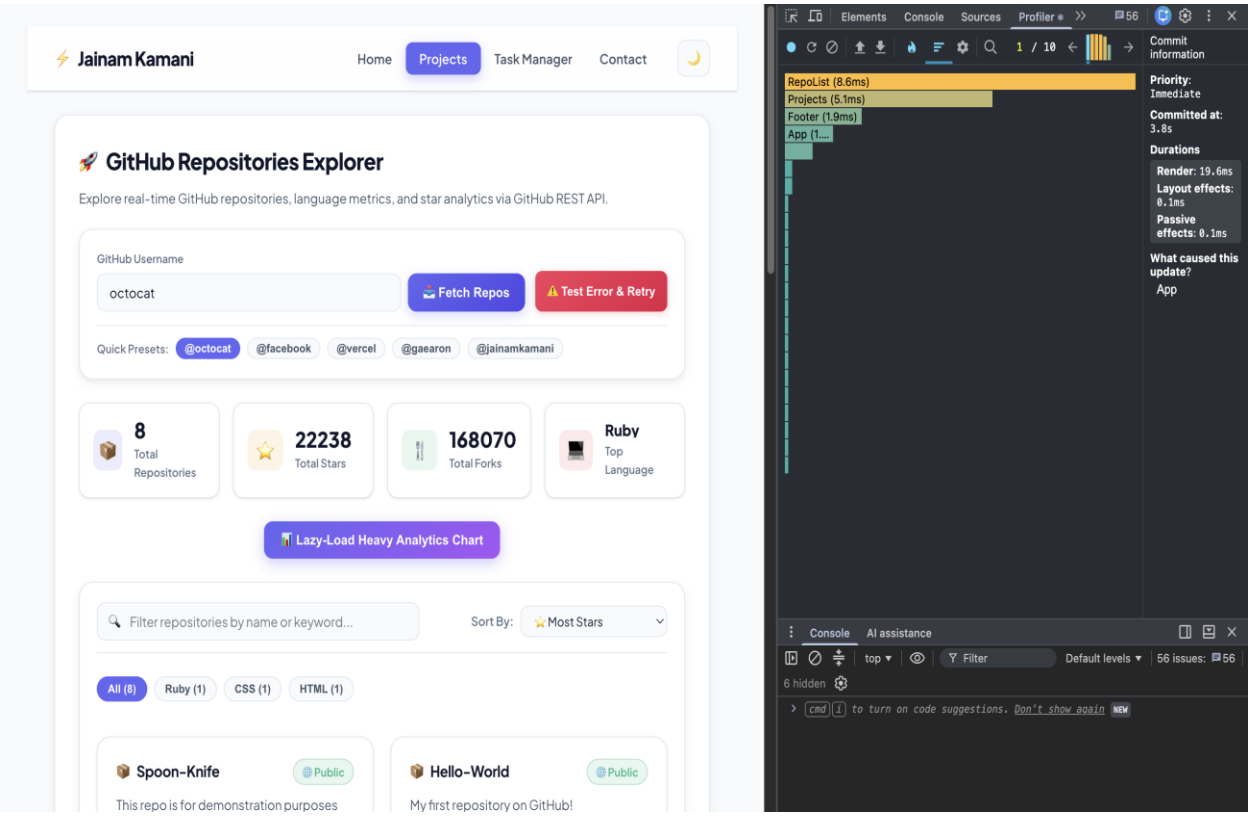

Jainam Kamani
Home
Projects
Task Manager
Contact
Get In Touch & Let's Connect
Have a question, project inquiry, or internship opportunity? Send a message below or reach out directly.
Send a Direct Message
Your Name *
e.g. Jane Doe
Your Email Address *
e.g. jane@example.com
Subject / Purpose
Project Inquiry
Message *
Write your detailed message here...
Send Message Now
Elements
Console
Sources
Network
Contact.jsx
200
script
App.jsx:12
34.9 kB
6 ms
1 requests
34.9 kB transferred
34.6 kB resources
Download the React DevTools for a better development experience:
https://react.dev/link/react-devtools


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GitHub Repositories Explorer
Explore real-time GitHub repositories, language metrics, and star analytics via GitHub REST API.
GitHub Username
octocat
Fetch Repos
Test Error & Retry
Quick Presets:
@octocat
@facebook
@vercel
@gaearon
@jainamkamani
0
Total Repositories
0
Total Stars
0
Total Forks
N/A
Top Language
Lazy-Load Heavy Analytics Chart
Filter repositories by name or keyword...
Sort By:
Most Stars
Fetching repositories for '@octocat' from GitHub API...
Elements
Console
Sources
Network
0 / 2 requests
0.0 kB / 0.0 kB transferred
0.0 kB / 0.0 kB resources
Download the React DevTools for a better development experience:
https://react.dev/link/react-devtools


Jainam Kamani
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Open for IT & Software Engineering Internships
Student Portfolio
Hello, I'm Jainam Kamani
Information Technology Student & Web Developer
Charotar University of Science and Technology (CHARUSAT)
B.Tech Information Technology
About Me
I am an Information Technology student at Charotar University of Science and Technology, originally from Junagadh, Gujarat. I am passionate about building modern, interactive web applications using technologies like React, JavaScript, and Vite, and constantly expanding my software development skills.
7.7 / 10
Cumulative CGPA
2
Projects Built
CHARUSAT
University
Junagadh
Hometown (Gujarat)
Elements
Console
Sources
Lighthouse
http://localhost:5173/projects
55
88
100
82
Performance
Accessibility
Best Practices
SEO
55
Performance
Values are estimated and may vary. The performance score is calculated directly from these metrics. See calculator.
0–49
50–89
90–100
Console
AI assistance
Download the React DevTools for a better development experience:
https://react.dev/link/react-devtools


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Task Manager
Contact
Application Performance & Bundle Analytics
JavaScript payload size across development phases
Bundle Size
Load Speed
Memory Heap
Unoptimized Peak
840 KB
P7 Monolith Build
Optimized Current
210 KB
-75%
P8 Lazy Chunked
Optimization Strategy
React.lazy() Code-Split
On-Demand Module Chunking
Hover over any bar below to view detailed breakdown and architectural release metrics.
480
520
590
650
710
780
840
210
P1
P2
P3
P4
P5
P6
P7
P8
Lazy Chunking Active
Baseline Monolith
Code Splitting via React.lazy()
-60% Initial Payload
Elements
Console
Sources
Network
Name
Status
Type
Initiator
Size
Time
AnalyticsChart.jsx
200
script
Projects.jsx:8
71.5 kB
9 ms
1 / 3 requests
71.5 kB / 82.1 kB transferred
71.2 kB / 164 kB resources
Console
AI assistance
Download the React DevTools for a better development experience:
https://react.dev/link/react-devtools

```
lazyWithMinDelay.js Practical8/... U X | AnalyticsChart.jsx M | LoadingFallback.jsx | lazyWithMinDelay.js
Practical8 > src > utils > lazyWithMinDelay.js > ...
1  import { lazy } from 'react';
2
3  /**
4   * Lazy loading helper with a configurable minimum delay.
5   * Prevents flickering of loading fallback states on fast network connections.
6   *
7   * @param {Function} importFn - Dynamic import function, e.g. () => import('./pages/Projects')
8   * @param {number} minDelayMs - Minimum duration to show fallback in milliseconds (default: 300ms)
9   * @returns {React.LazyExoticComponent}
10  */
11 export function lazyWithMinDelay(importFn, minDelayMs = 300) {
12   return lazy(() =>
13     Promise.all([
14       importFn(),
15       new Promise((resolve) => setTimeout(resolve, minDelayMs))
16     ]).then(([moduleExports]) => moduleExports)
17   );
18 }
19
20 export default lazyWithMinDelay;
```



Remember: React.lazy() loads the component when needed, while Suspense displays the fallback UI during loading.

KEY QUESTIONS / ANALYSIS:

1) What is Lazy Loading in React?

Lazy loading is a technique that loads a component only when it is required, rather than loading all components at the beginning. In React, we use React.lazy() to implement lazy loading.

2) What is Code Splitting in React?

Code splitting divides the application's JavaScript bundle into smaller chunks. These chunks can be loaded when needed, reducing the amount of JavaScript loaded initially.

3) What is the purpose of Suspense in React?

Suspense displays a fallback UI, such as *"Loading..."*, while a lazy-loaded component is being downloaded.

Important code example:

```
import { lazy, Suspense } from 'react';

const Projects = lazy(() => import('./Projects'));

function App() {
  return (
    <Suspense fallback={<div>Loading...</div>}>
      <Projects />
    </Suspense>
  );
}

export default App;
```

GITHUB REPOSITORY:

https://github.com/9081495207/ITUE301-Advanced-Web-Development-Frameworks